Cognitive Modeling

Complete Lecture Notes

Week 16: Final Project Presentations and Good Modeling Practice

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Chapter 1

Applications in Cognitive Science

1.1 Week 16: Final Project Presentations and Good Modeling Practice

> Course wrap-up · Lecture slot: synthesis + first presentations · Lab slot: presentations and peer feedback

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1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

English	فارسی	توضیح کوتاه
peer feedback	بازخورد همتا	نقد سازنده‌ی کار هم‌کلاسی‌ها
fit / fitting	برازش	تنظیم پارامترها تا مدل با داده جور شود
parameter recovery	بازیابی پارامتر	آیا برازش، پارامترهای معلوم شبیه‌سازی را برمی‌گرداند
reproducibility	بازتولیدپذیری	امکان به دست آوردن دوباره‌ی همان نتایج
simulation	شبیه‌سازی	تولید داده‌ی مصنوعی با اجرای مدل
estimate / estimator	برآوردگر / برآورد	روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
likelihood	درست‌نمایی	احتمال داده به‌عنوان تابعی از پارامترها
Bayes factor (BF)	عامل بیز (Bayes factor)	نسبت درست‌نمایی‌های حاشیه‌ای دو مدل

English	فارسی	توضیح کوتاه
bounds / constraints	قیدها / کران‌ها	محدودیت‌های مجاز برای مقدار پارامترها
posterior	توزیع پسین (posterior)	باور درباره‌ی پارامتر بعد از دیدن داده
hierarchical model	مدل سلسله‌مراتبی (چندسطحی)	پارامترهای فردی از یک توزیع گروهی می‌آیند
adequacy check	بررسی کفایت مدل	آیا مدل ویژگی‌های اصلی داده را بازتولید می‌کند
posterior predictive check (PPC)	بررسی پیش‌بینی پسین	مقایسه‌ی داده‌ی شبیه‌سازی‌شده از پسین با داده‌ی واقعی
cross-validation (CV)	اعتبارسنجی متقابل (cross-validation)	برازش روی بخشی از داده و آزمون روی بخش کنارگذاشته
AIC / BIC	بیزی / معیار اطلاعاتی آکائیکه	برازش به‌علاوه‌ی جریمه برای تعداد پارامتر
research design	طرح آزمایش	شرایط، تعداد کوشش‌ها و نمونه‌ها
model recovery	بازیابی مدل	آیا مقایسه‌ی مدل، مدل مولد را درست انتخاب می‌کند
drift-diffusion model (DDM)	مدل drift-diffusion (رانش–انتشار)	انباشت پیوسته‌ی شواهد بین دو مرز
preregistration	پیش‌ثبت (preregistration)	ثبت برنامه‌ی تحلیل پیش از دیدن داده
prior	توزیع پیشین (prior)	باور درباره‌ی پارامتر پیش از دیدن داده
random seed	seed تصادفی	عدد شروع مولد اعداد تصادفی برای تکرارپذیری
large language model (LLM)	مدل بزرگ زبانی	شبکه‌ی عصبی‌ای که کلمه‌ی بعدی متن را پیش‌بینی می‌کند
accuracy	دقت	نسبت پاسخ‌های درست

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Communicate a complete modeling study (question, model, fitting, evaluation, recovery, interpretation) clearly and within a time limit.
2. Critique other modeling studies constructively, using the course's evaluation criteria.
3. Apply a checklist of good modeling practice (reproducibility, transparency, robustness) to their own project report.
4. Integrate the methods of the course into one coherent workflow and explain when to use which.

1.1.3 Motivation & big picture

A model is only as useful as others' ability to understand, check and build on it. Over 15 weeks, every lab has followed the same arc: formalize → simulate → fit → evaluate → recover → interpret. This week you show that arc in your own project, and you practice the other half of scientific modeling: reading someone else's model critically but fairly. The session ends with the practices that make modeling work trustworthy and reusable: reproducible code, transparent reporting, honest statements of limitations.

1.1.4 Lecture notes

In plain words The last week has no new method. Students present their projects, and the course is summarized as one workflow: formalize a theory as a model, simulate it to see what it predicts, fit it to data, evaluate it (does it fit, and does it beat alternatives?), check that the method can recover the truth at this design, and interpret the result in psychological terms, including what cannot be concluded. The checklist of good practice turns these steps into concrete things to verify in every report: reproducible code, reported uncertainty, adequacy checks, recovery and honest limitations.

Suggested timing (lecture slot, 60 min): course synthesis 15 · good-practice checklist 15 · first presentations 30.

1. The course in one table

Question you face	Tools	Week
How do I turn a theory into a model?	Five components; simulation; analytic checks	1–2
How do I estimate parameters?	Least squares; maximum likelihood; optimization (grid, Nelder–Mead, L-BFGS-B), multi-start, bounds	3–4
How certain are the estimates?	Hessian SEs, profile likelihood, bootstrap; posterior intervals	5, 9
How do I handle many participants?	Individual fits; aggregation artifacts; hierarchical models	5, 13
Is the model adequate?	Residuals, G^2, posterior predictive checks	2–4, 10
Which model is better?	CV, LRT, AIC/BIC, Bayes factors	6–7, 10
Can my design answer the question?	Parameter recovery, model recovery	7

Question you face	Tools	Week
How do I get a posterior for complex models?	Grid → Metropolis–Hastings → Gibbs → NUTS (PyMC); $\hat{R}$, ESS, divergences	9–13
What do models look like in different fields?	SDT, forgetting, practice, GCM, DDM/EZ, RL, ideal observers	2, 5, 14–15

For an engineer Looking back, most of the course maps onto tools you already know, and the table below can serve as your own summary when answering students' questions:

Course topic	Engineering counterpart
Random walk / DDM (Weeks 1, 14)	Sequential detection (SPRT), first-passage times
SDT, ROC (Week 2)	Binary detection, P_D / P_{FA}, ROC
Least squares, optimizers (Week 3)	LS estimation, gradient methods, Gauss–Newton
Maximum likelihood, G^2, LRT (Weeks 4, 7)	ML estimation, goodness-of-fit tests, GLRT
Hessian SEs, bootstrap (Week 5)	Fisher information / CRLB, Monte Carlo performance evaluation
CV, AIC, BIC (Weeks 6–7)	Model-order selection (AIC, MDL), validation data
Parameter/model recovery (Week 7)	Testing estimators on synthetic data with known ground truth
Bayesian inference, BF (Weeks 9–10)	MAP/MMSE estimation, Bayesian detection
MCMC, $\hat{R}$, ESS (Weeks 11–12)	Monte Carlo simulation, Markov processes, correlated-sample statistics
Hierarchical models (Week 13)	Empirical Bayes, James–Stein shrinkage
Delta rule, cue combination (Week 15)	LMS adaptation, MRC / Kalman sensor fusion

What is genuinely different in cognitive modeling: the "system" is a human being, the data are small and noisy, and the parameters themselves are the scientific findings, so interpretation, uncertainty and recovery matter more than raw performance.

2. A checklist of good modeling practice

Before fitting - [ ] The research question is stated in terms of model parameters or model comparisons. - [ ] Candidate models (including at least one simpler alternative) are specified *before* looking at the data, ideally preregistered. - [ ] Priors (if Bayesian) are justified and checked with prior predictive simulation. - [ ] Parameter and model recovery have been checked at the actual design.

Fitting and evaluation - [ ] The likelihood matches the data type (binomial for counts, sequential for learning, ...). - [ ] Optimization uses multi-start and sensible bounds, and fits are checked visually;

for MCMC, $\hat{R}$, ESS and divergences are reported. - [ ] Absolute fit (residuals, G^2 or PPC) *and* relative fit (CV/IC/BF) are reported. - [ ] Uncertainty is reported for every interpreted parameter.

Reporting and reproducibility - [ ] All code and (where allowed) data are shared; random seeds are fixed. - [ ] Software versions are recorded (e.g., `condaenvexport>environment.yml`). - [ ] One command or notebook regenerates every number and figure in the report. - [ ] Limitations are stated: what the design cannot distinguish, which parameters are not recoverable, which assumptions were not tested.

Recording the computing environment (include in every project notebook):

Code · کد: `code/week16.py`, cell 1 · سلول 1

3. Presentation format

- Length: 8 minutes (individual) or 10 minutes (pairs), plus 4 minutes of questions. Strict time-keeping.
- Suggested structure (one slide each, roughly):
    1. The question and why it matters.
    2. The model(s): equations and the psychological meaning of every parameter.
    3. Data and design.
    4. Fitting method and a fit figure (data + predictions).
    5. Evaluation: model comparison *and* an adequacy check.
    6. Recovery: can the design answer the question?
    7. Interpretation and limitations.
- Pairs: both members present, and the Bayesian/MCMC or hierarchical component must be shown.
- Schedule: posted one week in advance; the order is drawn at random. Sessions run across the lecture and lab slots (about 15–18 presentations in 3 hours, depending on enrolment).

4. Giving and receiving feedback

Each student completes a peer-feedback form for three assigned presentations (below). Feedback is anonymous to the presenter, visible to the instructor, and counts toward participation (lab check-off). Good feedback is specific ("the recovery plot shows β is not identifiable, so the group difference in β should not be interpreted") rather than general ("nice talk"), and constructive (it suggests a concrete improvement).

Peer-feedback form

Item	Rating (1–4)	Comment
The question was clear and the model matched it		

Item	Rating (1–4)	Comment
The model's parameters were explained in psychological terms		
The fit was shown and evaluated (adequacy + comparison)		
Recovery or identifiability was addressed		
Conclusions followed from the results, and limitations were stated		
One thing that was particularly well done		
One concrete suggestion for the written report		

For the instructor

- Grade with the rubric, comment with the checklist. During presentations, keep the good-practice checklist next to you and note one strength and one concrete improvement per team; this makes the written feedback fast and fair.
- Questions to ask presenters (they map onto the course): Which parameter answers your question? How do you know it is recoverable? Does the model fit in absolute terms, not only better than the alternative? What can your design *not* tell us?
- Use the worked critique (fictional Q-learning study) as a model for how students should give feedback: specific, tied to a week of the course, and constructive.
- Close the course with the one-table summary. Ask students which row they found hardest, and point them to the corresponding week for exam revision.

1.1.5 Worked example: critiquing a modeling study

Summary of a (fictional) study. *"We fitted a Q-learning model to 30 patients and 30 controls who performed 60 trials of a two-armed bandit task with stable reward probabilities. Each participant was fitted by MLE. Patients had significantly lower learning rates ($t(58) = 2.4$, $p = .02$). We conclude that the disorder impairs reward learning."*

Critique using the checklist.

1. Recovery at this design (Week 7, Week 15 lab). With 60 trials and *stable* reward probabilities, the learning rate is weakly identifiable, and α and β trade off. Without a recovery study, a

difference in $\hat{\alpha}$ could reflect a difference in choice noise (β) or just estimation noise. *Suggestion:* report parameter recovery at this design, and include reversals.

2. Model comparison (Week 15, Part D). Did Q-learning beat simpler strategies (WSLS, perseveration) in both groups? If patients are better described by a different strategy, comparing their α with controls' is not meaningful. *Suggestion:* report AIC/BIC or CV per group, with model recovery.
3. Second-stage t-test on point estimates (Weeks 5, 13). It ignores the uncertainty of individual estimates, which is large with 60 trials. *Suggestion:* a hierarchical model with a group effect on α, reporting its posterior.
4. Adequacy (Weeks 4, 10). No posterior predictive check or learning-curve plot is shown. Does the model reproduce each group's learning curve?
5. Interpretation. "Impairs reward learning" is a strong process claim. A lower $\hat{\alpha}$ is also compatible with more exploration, lower motivation (lower β), or different priors on initial values. *Suggestion:* temper the conclusion, or add a design that separates these accounts.

Overall: a reasonable first study whose conclusion is not yet supported. Three additions (recovery, model comparison, hierarchical estimation) would make it convincing. Notice that each point maps directly onto one week of this course.

1.1.6 Lab (2 hours): presentations and peer feedback

Goals: present your project; give written feedback on three presentations; note the feedback you receive for your final report.

Logistics: - Upload slides (PDF) to the course page before the session starts. - Bring your laptop with your project notebook in case of questions about the code. - Fill in the peer-feedback forms during the session (paper or online form) and submit them by the end of the day.

After the presentations: the instructor gives brief oral feedback to each team. Written feedback (instructor + peers) is returned within 3 days so it can be used in the final report.

1.1.7 Connection to AI & LLMs

Good practice when AI is part of your modeling work. AI tools now enter cognitive modeling in three roles, and each needs the course's habits of checking: 1. As a coding assistant. Generated code can look right and be wrong. Test it the modeling way: simulate data with known parameters and check that the code recovers them (parameter recovery as a unit test). 2. As a "simulated participant". Some studies give LLMs psychology experiments and compare the answers with human data, or even propose LLM-based models of human behavior (e.g., Binz & Schulz, 2023; Dillion et al., 2023 [verify]). Evaluate such claims like any model: generalization to new tasks, comparison with simpler

cognitive models, mimicry, and interpretability (Week 15). 3. As an object of study. If you test an LLM, record the model name and version, the date, the exact prompts, the temperature and the number of samples. Without these, the study cannot be reproduced.

The code below tests a pre-made SDT function "written by an assistant". It looks plausible, and it even passes the case $c = 0$, but a recovery test shows that it gets the sign of c wrong.

Code · کد: `code/week16.py`, cell 2 · سلول 2

Expected output:

```
true (d', c) = (1.5, 0.3) -> recovered (1.50, -0.30)  FAIL
true (d', c) = (1.0, -0.4) -> recovered (1.00, 0.40)  FAIL
true (d', c) = (2.0, 0.0) -> recovered (2.00, 0.00)  OK
```

A liberal observer would be reported as conservative, which would reverse the conclusion of a bias study. The test with $c = 0$ alone would not have caught it.

Try it (no API needed). - ⋆ Fix the function and rerun the tests. - ⋆⋆ Write a similar recovery test for the EZ-diffusion function of Week 14, using the simulator from the same lab. - ⋆⋆ For your own project: add one paragraph to the reproducibility statement that describes how AI tools were used and how their output was verified (see `assessments/project_brief.md`, section 6).

Optional reading: Binz, M., & Schulz, E. (2023). Using cognitive psychology to understand GPT-3. *PNAS, 120*(6), e2218523120. Dillion, D., Tandon, N., Gu, Y., & Gray, K. (2023). Can AI language models replace human participants? *Trends in Cognitive Sciences, 27*, 597–600 [verify].

1.1.8 Common mistakes & misconceptions (in project presentations and reports)

1. Too much on the method, too little on the meaning. Every parameter and every result needs a psychological interpretation.
2. Showing a fit statistic without a fit figure. Always plot data against predictions.
3. Model comparison without an adequacy check, or vice versa.
4. Interpreting parameters that were not shown to be recoverable.
5. Overclaiming: "the model is correct", "the mind is Bayesian". State what the evidence supports and what it does not.
6. Irreproducible results: missing seeds, unrecorded package versions, notebook cells run out of order.
7. Running over time. Practice the talk; one idea per slide.

1.1.9 Check your understanding (integrative review questions for the final exam)

1. A colleague has accuracy and RT data from a speed/accuracy manipulation and wants to know whether the manipulation affects caution or evidence quality. Outline a full modeling plan, naming the model, the fitting method, the evaluation, and the recovery check. Answer Model: drift-diffusion (EZ-diffusion for a first pass, or the full DDM by MLE / hierarchical Bayes). Fit per participant and condition. Compare models in which the manipulation affects only a, only v, or both (AIC/BIC or a hierarchical model with condition effects; PPC on RT quantiles). Recovery: simulate data at the actual design with each model, and check that parameters and the "which parameter changed" conclusion are recovered. Interpret the condition effects on a and v with uncertainty.

2. When would you choose a Bayes factor over cross-validation for comparing two models, and vice versa? Answer Bayes factor: when the question is which model is more probable given the data (evidence, including support for a null), when meaningful priors are available, and with a prior sensitivity analysis. CV: when the goal is predictive accuracy for new data of a specific kind, priors are hard to justify, or the BF is too sensitive to the priors.

3. Explain how a parameter can be estimated very precisely (narrow CI or posterior) and yet be meaningless. Answer Precision only reflects the curvature of the likelihood under the assumed model. If the model is misspecified (fails PPC/G^2), if the parameter trades off with another so that its meaning depends on an arbitrary constraint, or if the optimizer or sampler failed (local minimum, bound, non-convergence), the precise estimate does not measure the intended construct.

4. Why do hierarchical models often give better individual estimates than separate individual fits? Answer They share information across participants through the group distribution, shrinking noisy individual estimates toward the group mean in proportion to their uncertainty. This reduces estimation variance more than it adds bias, so estimates are on average closer to the truth.

5. Name three different ways in which this course used simulation, and why each matters. Answer (i) Exploring a model's predictions before fitting (what can and can't it produce). (ii) Parameter and model recovery (can the design and method answer the question). (iii) Bootstrap and predictive checks (uncertainty and adequacy). Further: Monte Carlo and MCMC for computing posteriors.

6. A model has the best AIC among four candidates but a posterior predictive check shows a systematic misfit at long delays. What do you report and conclude? Answer Report both. The model is the best of the candidates for prediction, but it is not adequate in absolute terms: none of the candidates captures the long-delay behavior. Conclusions should be restricted accordingly, and a model revision targeting that feature is the next step.

1.1.10 Assignment

Final project: written report and code

- Due: one week after this session, and in any case before the final exam (exact date on the course page).

- Submission: report (PDF, max. 10 pages for individuals, 14 for pairs, excluding references and appendix) plus a code archive or repository link that reproduces all results.
- Grading: see the project brief and rubric (`assessments/project_brief.md`). The report and code are worth 3 of the 6 project points; the proposal (1) and presentation (2) are already graded.
- Use the feedback from today: the rubric rewards visible responses to presentation feedback.

Final exam

Written, cumulative (emphasis on Chapters 4–6), formula sheet provided, no laptops; worth 3 of 20 points. See `assessments/final_exam.md` for the format and a sample paper.

1.1.11 Further reading

- Farrell & Lewandowsky (2018): the chapter on using models in psychology (the role of models in theory development and common pitfalls) [verify chapter number].
- Lee & Wagenmakers (2014): browse the case studies in Part IV for ideas on how to extend your project.
- Martin, Kumar & Lao (2021), chapter 9. End-to-end Bayesian workflows, a good model for structuring a report.
- *Optional, outside the course list:* Wilson, R. C., & Collins, A. G. E. (2019). Ten simple rules for the computational modeling of behavioral data. *eLife, 8*, e49547.

Thank you for a great course, and keep modeling!